jet. When they compared the clocks flown on the plane with identical clocks left stationary on the ground, they found that less time had elapsed on the moving clocks. The difference was tiny—a few hundred billionths of a second—but it was precisely in accord with Einstein's discoveries. You can't get much more nuts-and-bolts than that.

In 1908, word began to spread that newer, more refined experiments were finding evidence for the aether.[6] If that had been so, it would have meant that there was an absolute standard of rest and that Einstein's special relativity was wrong. On hearing this rumor, Einstein replied, "Subtle is the Lord, malicious He is not." Peering deeply into the workings of nature to tease out insights into space and time was a profound challenge, one that had gotten the better of everyone until Einstein. But to allow such a startling and beautiful theory to exist, and yet to make it irrelevant to the workings of the universe, that would be malicious. Einstein would have none of it; he dismissed the new experiments. His confidence was well placed. The experiments were ultimately shown to be wrong, and the luminiferous aether evaporated from scientific discourse.

But What About the Bucket?

This is certainly a tidy story for light. Theory and experiment agree that light needs no medium to carry its waves and that regardless of the motion of either the source of light or the person observing, its speed is fixed and unchanging. Every vantage point is on an equal footing with every other. There is no absolute or preferred standard of rest. Great. But what about the bucket?

Remember, while many viewed the luminiferous aether as the physical substance giving credibility to Newton's absolute space, it had nothing to do with *why* Newton introduced absolute space. Instead, after wrangling with accelerated motion such as the spinning bucket, Newton saw no option but to invoke some invisible background stuff with respect to which motion could be unambiguously defined. Doing away with the aether did not do away with the bucket, so how did Einstein and his special theory of relativity cope with the issue?